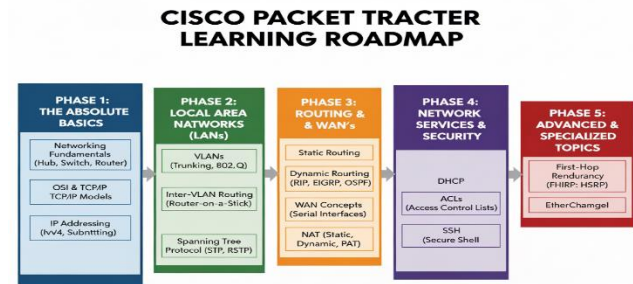


Complete CCNA Study Guide with Lab Tasks

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Phase 1: The Absolute Basics

1.1 Networking Fundamentals

Key Concepts

Network Devices:

- **Hub:** Layer 1 device, creates single collision domain, broadcasts to all ports
- **Switch:** Layer 2 device, learns MAC addresses, creates separate collision domains per port
- **Router:** Layer 3 device, routes between networks using IP addresses, separates broadcast domains

Collision Domain vs Broadcast Domain:

- **Collision Domain:** Area where data collisions can occur (hubs create one large collision domain)
- **Broadcast Domain:** Area where broadcast frames are propagated (routers separate broadcast domains)

Lab Task 1.1: Basic Network Connectivity

Topology:

PC1 ---- Switch ---- PC2

Configuration Steps:

Physical Connections:

Connect PC1 to Switch Fa0/1

Connect PC2 to Switch Fa0/2

Use copper straight-through cables

PC Configuration:

PC1:
IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

PC2:
IP Address: 192.168.1.20
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Testing:

```
PC1> ping 192.168.1.20  
PC1> arp -a
```

Expected Results:

- Ping should be successful
- ARP table should show PC2's MAC address

1.2 OSI & TCP/IP Models

OSI Model (7 Layers)

Layer	Name	Function	Devices/Protocols
7	Application	User interface	HTTP, FTP, SMTP
6	Presentation	Data formatting, encryption	SSL, JPEG, ASCII
5	Session	Session management	NetBIOS, SQL
4	Transport	End-to-end delivery	TCP, UDP
3	Network	Logical addressing, routing	IP, ICMP, Routers
2	Data Link	Physical addressing, switching	Ethernet, Switches
1	Physical	Transmission medium	Cables, Hubs

TCP/IP Model (4 Layers)

Layer	OSI Equivalent	Function
Application	5-7	Application services
Transport	4	End-to-end communication
Internet	3	Routing
Network Access	1-2	Physical network access

Lab Task 1.2: Protocol Analysis

Objective: Understand how different protocols work at different layers

Steps:

1. Set up the same topology as Task 1.1
2. Use Packet Tracer simulation mode
3. Send a ping from PC1 to PC2
4. Analyze the packet at each layer:
 - Layer 1: Electrical signals
 - Layer 2: Ethernet frame (MAC addresses)
 - Layer 3: IP packet (IP addresses)
 - Layer 4: ICMP message

Commands for Analysis:

```
PC1> ping 192.168.1.20
PC1> tracert 192.168.1.20
```

1.3 IP Addressing (IPv4/IPv6)

IPv4 Addressing

Address Classes:

- Class A: 1.0.0.0 to 126.0.0.0 (/8)
- Class B: 128.0.0.0 to 191.255.0.0 (/16)
- Class C: 192.0.0.0 to 223.255.255.0 (/24)

Private Address Ranges:

- Class A: 10.0.0.0/8
- Class B: 172.16.0.0/12
- Class C: 192.168.0.0/16

Special Addresses:

- Loopback: 127.0.0.0/8
- APIPA: 169.254.0.0/16
- Multicast: 224.0.0.0/4

Subnetting

CIDR Notation:

- /24 = 255.255.255.0
- /25 = 255.255.255.128
- /26 = 255.255.255.192
- /27 = 255.255.255.224

Subnetting Formula:

- Number of subnets = 2^n (where n = borrowed bits)
- Hosts per subnet = $2^h - 2$ (where h = host bits)

Lab Task 1.3: Subnetting Practice

Scenario: Subnet 192.168.1.0/24 into 4 subnets

Solution:

- Borrowed bits: 2 ($2^2 = 4$ subnets)
- New subnet mask: /26 (255.255.255.192)
- Subnets:
 - 192.168.1.0/26 (192.168.1.1-62)
 - 192.168.1.64/26 (192.168.1.65-126)
 - 192.168.1.128/26 (192.168.1.129-190)
 - 192.168.1.192/26 (192.168.1.193-254)

Topology:

PC1 (192.168.1.10/26) ---- Switch ---- PC2 (192.168.1.70/26)

Testing:

```
PC1> ping 192.168.1.70
```

This should fail because they're in different subnets without a router.

IPv6 Addressing

Address Types:

- **Unicast:** One-to-one communication
- **Multicast:** One-to-many communication

- **Anycast:** One-to-nearest communication

Address Format:

- 128-bit address
- 8 groups of 4 hexadecimal digits
- Example: 2001:db8:85a3::8a2e:370:7334

Special IPv6 Addresses:

- Loopback: ::1/128
 - Link-local: fe80::/10
 - Unique local: fc00::/7
-

Phase 2: Local Area Networks (LANs)

2.1 Ethernet & Switching

Ethernet Standards

Standard	Speed	Cable Type
10BASE-T	10 Mbps	Cat 3 UTP
100BASE-TX	100 Mbps	Cat 5 UTP
1000BASE-T	1 Gbps	Cat 5e/6 UTP
10GBASE-T	10 Gbps	Cat 6a UTP

Switch Operation

MAC Address Table:

- Switches learn MAC addresses dynamically
- Default aging time: 300 seconds
- CAM table stores MAC-to-port mappings

Frame Processing:

1. **Learn:** Source MAC address
2. **Forward:** Based on destination MAC
3. **Flood:** If destination unknown
4. **Filter:** If source and destination on same port

Lab Task 2.1: Switch MAC Address Learning

Topology:

```
PC1 ---- Fa0/1 [Switch] Fa0/2 ---- PC2
          |
          Fa0/3
          |
          PC3
```

Switch Configuration:

```
Switch> enable
Switch# show mac address-table
Switch# clear mac address-table dynamic
```

Testing Sequence:

1. Clear MAC table
2. Send ping from PC1 to PC2
3. Check MAC table after each step
4. Observe learning process

Expected Commands:

```
PC1> ping 192.168.1.20
Switch# show mac address-table
Switch# show mac address-table dynamic interface fa0/1
```

2.2 VLANs (Virtual LANs)

VLAN Concepts

Benefits:

- Logical segmentation
- Improved security
- Broadcast domain control
- Simplified management

VLAN Types:

- **Data VLAN:** User traffic
- **Voice VLAN:** VoIP traffic
- **Management VLAN:** Switch management
- **Native VLAN:** Untagged traffic on trunk

VLAN Configuration

Basic VLAN Commands:

```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config)# vlan 20
Switch(config-vlan)# name Engineering

Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10

Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
```

Lab Task 2.2: VLAN Configuration

Topology:

```
PC1(VLAN 10) ---- Fa0/1 [Switch1] Fa0/24 ---- Fa0/24 [Switch2] Fa0/1 ----
PC3(VLAN 10)
PC2(VLAN 20) ---- Fa0/2                                Fa0/2 ----
PC4(VLAN 20)
```

Switch1 Configuration:

```
Switch1(config)# vlan 10
Switch1(config-vlan)# name Sales
Switch1(config)# vlan 20
Switch1(config-vlan)# name Engineering

Switch1(config)# interface fa0/1
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 10

Switch1(config)# interface fa0/2
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 20

Switch1(config)# interface fa0/24
Switch1(config-if)# switchport mode trunk
```

Switch2 Configuration:

```
Switch2(config)# vlan 10
Switch2(config-vlan)# name Sales
Switch2(config)# vlan 20
Switch2(config-vlan)# name Engineering

Switch2(config)# interface fa0/1
Switch2(config-if)# switchport mode access
Switch2(config-if)# switchport access vlan 10

Switch2(config)# interface fa0/2
Switch2(config-if)# switchport mode access
```

```
Switch2(config-if)# switchport access vlan 20
```

```
Switch2(config)# interface fa0/24
```

```
Switch2(config-if)# switchport mode trunk
```

Verification Commands:

```
Switch# show vlan brief
```

```
Switch# show interfaces trunk
```

```
Switch# show interfaces fa0/1 switchport
```

2.3 Trunking (802.1Q)

Trunking Concepts

802.1Q Frame Format:

- 4-byte VLAN tag inserted into Ethernet frame
- VLAN ID field: 12 bits (4094 possible VLANs)
- Native VLAN: Untagged traffic (default VLAN 1)

Trunk Negotiation Protocols:

- **DTP (Dynamic Trunking Protocol):** Cisco proprietary
- **Manual configuration:** Recommended for security

Trunking Configuration

Trunk Port Configuration:

```
Switch(config)# interface fa0/24
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# switchport trunk allowed vlan 10,20
```

```
Switch(config-if)# switchport trunk native vlan 99
```

DTP Modes:

- `switchport mode trunk`: Permanent trunk
- `switchport mode access`: Permanent access
- `switchport mode dynamic auto`: Trunk if neighbour desires
- `switchport mode dynamic desirable`: Actively negotiate trunk

Lab Task 2.3: Advanced Trunking

Objective: Configure trunk with specific VLANs and native VLAN

Configuration:

```
Switch1(config)# interface fa0/24
```

```
Switch1(config-if)# switchport trunk encapsulation dot1q
```



```
Switch1(config-if)# switchport mode trunk
Switch1(config-if)# switchport trunk allowed vlan 10,20,30
Switch1(config-if)# switchport trunk native vlan 99
Switch1(config-if)# switchport nonegotiate
```

Verification:

```
Switch# show interfaces fa0/24 trunk
Switch# show interfaces fa0/24 switchport
Switch# show dtp interface fa0/24
```

2.4 Spanning Tree Protocol (STP)

STP Concepts

Purpose:

- Prevent Layer 2 loops
- Provide redundancy
- Automatic failover

STP States:

1. **Blocking:** Receives BPDUs, blocks data
2. **Listening:** Processes BPDUs, no MAC learning
3. **Learning:** Builds MAC table, no forwarding
4. **Forwarding:** Normal operation
5. **Disabled:** Administratively down

Port Roles:

- **Root Port:** Best path to root bridge
- **Designated Port:** Sends BPDUs on segment
- **Blocked Port:** Prevents loops

STP Configuration

Root Bridge Configuration:

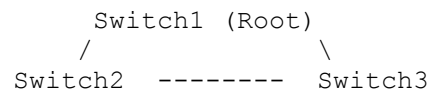
```
Switch(config)# spanning-tree vlan 1 priority 4096
Switch(config)# spanning-tree vlan 1 root primary
```

Port Priority Configuration:

```
Switch(config)# interface fa0/1
Switch(config-if)# spanning-tree port-priority 128
Switch(config-if)# spanning-tree cost 19
```

Lab Task 2.4: STP Convergence

Topology:



Configuration:

```
# Make Switch1 root bridge
Switch1(config)# spanning-tree vlan 1 priority 4096

# Verify STP topology
Switch# show spanning-tree
Switch# show spanning-tree summary
Switch# show spanning-tree interface fa0/1
```

Testing:

1. Verify port states
 2. Disconnect primary link
 3. Observe reconvergence
 4. Check new port states
-

2.5 VTP (VLAN Trunking Protocol)

VTP Concepts

VTP Modes:

- **Server:** Can create/modify/delete VLANs
- **Client:** Cannot modify VLAN database
- **Transparent:** Forwards VTP messages, independent database

VTP Requirements:

- Same VTP domain name
- Same VTP password (if configured)
- Trunk links between switches

VTP Configuration

Server Configuration:

```
Switch(config)# vtp domain COMPANY
Switch(config)# vtp mode server
Switch(config)# vtp password cisco123
Switch(config)# vtp version 2
```

Client Configuration:

```
Switch(config)# vtp domain COMPANY
Switch(config)# vtp mode client
Switch(config)# vtp password cisco123
```

Lab Task 2.5: VTP Domain Setup

Topology:

```
Switch1(Server) ---- Switch2(Client) ---- Switch3(Transparent)
```

Configuration Steps:

1. Configure VTP domain on all switches
2. Set modes appropriately
3. Create VLANs on server
4. Verify propagation to client
5. Verify transparent mode behavior

Verification:

```
Switch# show vtp status
Switch# show vtp counters
Switch# show vlan brief
```

Phase 3: Routing & WANs

3.1 Basic Router Configuration

Router Boot Process

1. **POST (Power-On Self Test)**
2. **Bootstrap program**
3. **IOS loading**
4. **Configuration loading**

Initial Router Configuration

Basic Setup:

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# enable password cisco
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password console
R1(config-line)# login
R1(config-line)# exit
```

```
R1(config)# line vty 0 4
R1(config-line)# password telnet
R1(config-line)# login
R1(config-line)# exit
R1(config)# service password-encryption
R1(config)# banner motd #Unauthorized Access Prohibited#
```

Interface Configuration:

```
R1(config)# interface gigabitethernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# description LAN Interface
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.1.1.1 255.255.255.252
R1(config-if)# description WAN Interface
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
```

Lab Task 3.1: Router Initial Configuration

Topology:

```
PC1 ---- Gi0/0 [R1] S0/0/0 ---- S0/0/0 [R2] Gi0/0 ---- PC2
192.168.1.0/24    10.1.1.0/30      192.168.2.0/24
```

R1 Configuration:

```
R1(config)# interface gi0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
```

```
R1(config)# interface s0/0/0
R1(config-if)# ip address 10.1.1.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
```

R2 Configuration:

```
R2(config)# interface gi0/0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
```

```
R2(config)# interface s0/0/0
R2(config-if)# ip address 10.1.1.2 255.255.255.252
R2(config-if)# no shutdown
```

Verification:

```
R1# show ip interface brief
R1# show running-config
R1# ping 10.1.1.2
R1# show cdp neighbors
```

3.2 Static Routing

Static Route Concepts

Route Types:

- **Directly Connected:** Interface networks
- **Static:** Manually configured
- **Dynamic:** Learned via routing protocols

Administrative Distance:

- Connected: 0
- Static: 1
- EIGRP: 90
- OSPF: 110
- RIP: 120

Static Route Configuration

Standard Static Route:

```
Router(config)# ip route destination_network subnet_mask next_hop_ip
Router(config)# ip route 192.168.2.0 255.255.255.0 10.1.1.2
```

Default Route:

```
Router(config)# ip route 0.0.0.0 0.0.0.0 next_hop_ip
```

Summary Route:

```
Router(config)# ip route 172.16.0.0 255.255.0.0 10.1.1.2
```

Lab Task 3.2: Static Routing Configuration

Topology:

```
PC1 ---- [R1] ---- [R2] ---- [R3] ---- PC2
    192.168.1.0/24 | 10.1.1.0/30 | 192.168.3.0/24
                  |             |
                  |             |
                192.168.2.0/24  10.1.1.4/30
```

R1 Static Routes:

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.1.1.2
R1(config)# ip route 192.168.3.0 255.255.255.0 10.1.1.2
R1(config)# ip route 10.1.1.4 255.255.255.252 10.1.1.2
```

R2 Static Routes:

```
R2(config)# ip route 192.168.1.0 255.255.255.0 10.1.1.1
R2(config)# ip route 192.168.3.0 255.255.255.0 10.1.1.6
```

R3 Static Routes:

```
R3(config)# ip route 192.168.1.0 255.255.255.0 10.1.1.5
R3(config)# ip route 192.168.2.0 255.255.255.0 10.1.1.5
R3(config)# ip route 10.1.1.0 255.255.255.252 10.1.1.5
```

Verification:

```
R1# show ip route
R1# show ip route static
PC1> tracert 192.168.3.10
```

3.3 Dynamic Routing (RIP, EIGRP, OSPF)

RIP (Routing Information Protocol)

RIP Characteristics:

- Distance vector protocol
- Hop count metric (max 15)
- Updates every 30 seconds
- Uses broadcast/multicast

RIP Configuration:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP Characteristics:

- Cisco proprietary (now open standard)
- Advanced distance vector
- Fast convergence
- Unequal cost load balancing

EIGRP Configuration:

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.1.1.0 0.0.0.3
Router(config-router)# no auto-summary
```

OSPF (Open Shortest Path First)

OSPF Characteristics:

- Link state protocol
- Fast convergence
- Hierarchical design with areas
- Cost metric based on bandwidth

OSPF Configuration:

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

Lab Task 3.3: Dynamic Routing Protocols

Topology for RIP:

PC1 ---- [R1] ---- [R2] ---- [R3] ---- PC2

RIP Configuration:

```
# On all routers
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network [directly_connected_networks]
Router(config-router)# no auto-summary
```

OSPF Configuration:

```
# On all routers
Router(config)# router ospf 1
Router(config-router)# network 0.0.0.0 255.255.255.255 area 0
```

Verification Commands:

```
Router# show ip route
Router# show ip protocols
Router# show ip rip database
Router# show ip ospf database
Router# show ip eigrp topology
```

3.4 WAN Technologies

WAN Connection Types

Serial Connections:

- Point-to-point links
- Frame Relay

- ATM
- MPLS

PPP (Point-to-Point Protocol):

```
Router(config)# interface serial 0/0/0
Router(config-if)# encapsulation ppp
Router(config-if)# ppp authentication chap
Router(config-if)# username remote_router password shared_secret
```

Frame Relay:

```
Router(config)# interface serial 0/0/0
Router(config-if)# encapsulation frame-relay
Router(config-if)# frame-relay map ip 192.168.1.2 102 broadcast
```

Lab Task 3.4: PPP Configuration

Topology:

R1 ---- PPP Link ---- R2

R1 Configuration:

```
R1(config)# username R2 password cisco123
R1(config)# interface s0/0/0
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication chap
R1(config-if)# ip address 10.1.1.1 255.255.255.252
R1(config-if)# no shutdown
```

R2 Configuration:

```
R2(config)# username R1 password cisco123
R2(config)# interface s0/0/0
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication chap
R2(config-if)# ip address 10.1.1.2 255.255.255.252
R2(config-if)# no shutdown
```

Verification:

```
R1# show ppp multilink
R1# debug ppp authentication
```

3.5 NAT (Network Address Translation)

NAT Types

Static NAT:

- One-to-one mapping

- Permanent translation

Dynamic NAT:

- Pool of public addresses
- First-come, first-served

PAT (Port Address Translation):

- Many-to-one using port numbers
- Most common implementation

NAT Configuration

Static NAT:

```
Router(config)# ip nat inside source static 192.168.1.10 203.0.113.10
Router(config)# interface gi0/0
Router(config-if)# ip nat inside
Router(config)# interface s0/0/0
Router(config-if)# ip nat outside
```

Dynamic NAT:

```
Router(config)# ip nat pool PUBLIC 203.0.113.1 203.0.113.10 netmask
255.255.255.0
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 pool PUBLIC
```

PAT Configuration:

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface s0/0/0 overload
Router(config)# interface gi0/0
Router(config-if)# ip nat inside
Router(config)# interface s0/0/0
Router(config-if)# ip nat outside
```

Lab Task 3.5: NAT/PAT Configuration

Topology:

```
PC1 ---- [Router] ---- Internet
PC2 ---- NAT ---- (ISP Router)
192.168.1.0/24 | 203.0.113.1/30
```

Router Configuration:

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface s0/0/0 overload

Router(config)# interface gi0/0
Router(config-if)# ip nat inside
```

```
Router(config)# interface s0/0/0
Router(config-if)# ip nat outside
```

Verification:

```
Router# show ip nat translations
Router# show ip nat statistics
Router# clear ip nat translation *
```

Phase 4: Network Services & Security

4.1 DHCP (Dynamic Host Configuration Protocol)

DHCP Process (DORA)

1. **Discover:** Client broadcasts DHCP discover
2. **Offer:** Server responds with IP offer
3. **Request:** Client requests specific IP
4. **Acknowledge:** Server confirms assignment

DHCP Configuration

Router as DHCP Server:

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)# ip dhcp pool LAN_POOL
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
Router(dhcp-config)# lease 7
```

DHCP Relay Agent:

```
Router(config)# interface gi0/0
Router(config-if)# ip helper-address 192.168.100.10
```

Lab Task 4.1: DHCP Server Configuration

Topology:

```
PC1 ---- [Router/DHCP Server] ---- [ISP Router]
PC2 ---- 192.168.1.1/24          ---- Internet
```

DHCP Configuration:

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)# ip dhcp pool INTERNAL
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# domain-name company.com
```

```
Router(dhcp-config)# lease 0 12 0
```

Client Configuration:

- Set PCs to obtain IP automatically
- Release and renew IP addresses

Verification:

```
Router# show ip dhcp binding
Router# show ip dhcp pool
Router# show ip dhcp conflict
PC> ipconfig /all
PC> ipconfig /release
PC> ipconfig /renew
```

4.2 Telnet & SSH

Telnet Configuration

Basic Telnet Setup:

```
Router(config)# line vty 0 4
Router(config-line)# password telnet123
Router(config-line)# login
Router(config-line)# transport input telnet
```

SSH Configuration

SSH Setup:

```
Router(config)# hostname R1
R1(config)# ip domain-name company.com
R1(config)# crypto key generate rsa
R1(config)# ip ssh version 2
R1(config)# username admin privilege 15 secret admin123
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
```

Lab Task 4.2: SSH Configuration

Objective: Configure secure remote access to router

Configuration:

```
Router(config)# hostname R1
R1(config)# ip domain-name lab.local
R1(config)# crypto key generate rsa general-keys modulus 1024
R1(config)# ip ssh time-out 120
R1(config)# ip ssh authentication-retries 3
R1(config)# username admin privilege 15 secret cisco123
R1(config)# line vty 0 4
```

```
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exec-timeout 10 0
```

Testing SSH Access:

```
PC> ssh -l admin 192.168.1.1
```

Verification:

```
R1# show ssh
R1# show ip ssh
R1# show users
R1# show login
```

4.3 Access Control Lists (ACLs)

ACL Types

Standard ACLs (1-99, 1300-1999):

- Filter based on source IP only
- Applied close to destination

Extended ACLs (100-199, 2000-2699):

- Filter based on source, destination, protocol, port
- Applied close to source

Named ACLs:

- More descriptive than numbered
- Allow editing of individual entries

Standard ACL Configuration

Numbered Standard ACL:

```
Router(config)# access-list 10 permit 192.168.1.0 0.0.0.255
Router(config)# access-list 10 deny 192.168.2.10 0.0.0.0
Router(config)# access-list 10 permit any

Router(config)# interface gi0/1
Router(config-if)# ip access-group 10 out
```

Named Standard ACL:

```
Router(config)# ip access-list standard SALES_FILTER
Router(config-std-nacl)# permit 192.168.10.0 0.0.0.255
Router(config-std-nacl)# deny any
```

```
Router(config)# interface gi0/0
Router(config-if)# ip access-group SALES_FILTER in
```

Extended ACL Configuration

Numbered Extended ACL:

```
Router(config)# access-list 101 deny tcp 192.168.1.0 0.0.0.255 192.168.2.0
0.0.0.255 eq 80
Router(config)# access-list 101 permit tcp any any eq 443
Router(config)# access-list 101 permit icmp any any
Router(config)# access-list 101 deny ip any any

Router(config)# interface gi0/0
Router(config-if)# ip access-group 101 in
```

Named Extended ACL:

```
Router(config)# ip access-list extended WEB_FILTER
Router(config-ext-nacl)# deny tcp 192.168.1.0 0.0.0.255 any eq 80
Router(config-ext-nacl)# permit tcp 192.168.1.0 0.0.0.255 any eq 443
Router(config-ext-nacl)# permit icmp any any echo-reply
Router(config-ext-nacl)# permit icmp any any unreachable
Router(config-ext-nacl)# deny ip any any log

Router(config)# interface gi0/0
Router(config-if)# ip access-group WEB_FILTER out
```

Lab Task 4.3: Comprehensive ACL Implementation

Topology:

```
Sales_PC ---- [Router] ---- Web_Server
(192.168.10.10) |      (192.168.20.10)
                |
Engineering_PC ---
(192.168.10.20)
```

Requirements:

1. Sales can access web server HTTP only
2. Engineering can access all services
3. Block all other traffic
4. Allow ICMP for troubleshooting

ACL Configuration:

```
Router(config)# ip access-list extended DEPARTMENT_POLICY
Router(config-ext-nacl)# remark Allow Sales HTTP access
Router(config-ext-nacl)# permit tcp host 192.168.10.10 host 192.168.20.10
eq 80
Router(config-ext-nacl)# remark Allow Engineering full access
Router(config-ext-nacl)# permit ip host 192.168.10.20 192.168.20.0
0.0.0.255
Router(config-ext-nacl)# remark Allow ICMP for troubleshooting
```

```
Router(config-ext-nacl)# permit icmp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config-ext-nacl)# remark Deny all other traffic
Router(config-ext-nacl)# deny ip any any log

Router(config)# interface gi0/0
Router(config-if)# ip access-group DEPARTMENT_POLICY out
```

Verification:

```
Router# show access-lists
Router# show access-lists DEPARTMENT_POLICY
Router# show ip interface gi0/0
```

Testing:

```
Sales_PC> ping 192.168.20.10
Engineering_PC> telnet 192.168.20.10
```

4.4 Additional Network Services

NTP (Network Time Protocol)

NTP Client Configuration:

```
Router(config)# clock timezone PST -8
Router(config)# clock summer-time PDT recurring
Router(config)# ntp server 129.6.15.28
Router(config)# ntp server 132.163.4.101
```

NTP Server Configuration:

```
Router(config)# ntp master 3
Router(config)# ntp source gi0/0
```

SNMP (Simple Network Management Protocol)

SNMPv2c Configuration:

```
Router(config)# snmp-server community public ro
Router(config)# snmp-server community private rw
Router(config)# snmp-server location "Network Closet A"
Router(config)# snmp-server contact "admin@company.com"
```

SNMPv3 Configuration:

```
Router(config)# snmp-server group ADMINS v3 priv
Router(config)# snmp-server user admin ADMINS v3 auth sha cisco123 priv aes 128 cisco456
Router(config)# snmp-server host 192.168.1.100 version 3 priv admin
```

Syslog Configuration

Syslog Setup:

```
Router(config)# logging 192.168.1.50
Router(config)# logging trap informational
Router(config)# logging source-interface gi0/0
Router(config)# service timestamps log datetime msec
Router(config)# logging buffered 4096 debugging
```

Lab Task 4.4: Network Services Integration

Objective: Configure multiple network services for network management

NTP Configuration:

```
Router(config)# ntp server pool.ntp.org
Router(config)# clock timezone EST -5
Router(config)# service timestamps log datetime msec
```

Syslog Configuration:

```
Router(config)# logging host 192.168.1.100
Router(config)# logging trap notifications
Router(config)# logging source-interface gi0/0
```

SNMP Configuration:

```
Router(config)# snmp-server community monitor ro
Router(config)# snmp-server location "Main Office Router"
Router(config)# snmp-server contact "Network Admin"
```

Verification:

```
Router# show ntp status
Router# show clock
Router# show logging
Router# show snmp
```

Phase 5: Advanced & Specialized Topics

5.1 Wireless LAN (Standards, Security)

Wireless Standards

Standard	Frequency	Max Speed	Range
----------	-----------	-----------	-------

802.11a	5 GHz	54 Mbps	35m
802.11b	2.4 GHz	11 Mbps	140m
802.11g	2.4 GHz	54 Mbps	140m

Standard Frequency Max Speed Range

802.11n	2.4/5 GHz	600 Mbps	250m
802.11ac	5 GHz	6.93 Gbps	35m
802.11ax	2.4/5 GHz	9.6 Gbps	35m

Wireless Security

Security Protocols:

- **WEP:** Deprecated, easily cracked
- **WPA:** TKIP encryption, PSK or Enterprise
- **WPA2:** AES encryption, most common
- **WPA3:** Enhanced security, SAE authentication

Wireless Controller Configuration

Basic Wireless Setup:

```
Controller(config)# wlan create 10 CORPORATE ssid CORP-WIFI
Controller(config-wlan)# security wpa psk set-key ascii 0 StrongPassword123
Controller(config-wlan)# security wpa akm psk
Controller(config-wlan)# no shutdown
```

```
Controller(config)# interface create VLAN10 10
Controller(config)# interface vlan 10 dhcp required
```

Lab Task 5.1: Wireless Network Configuration

Topology:

```
Laptop ~~~ Access Point ---- Switch ---- Router
           |               |           |
           Controller      DHCP Server Internet
```

Access Point Configuration:

```
AP(config)# interface dot11radio 0
AP(config-if)# ssid CORPORATE
AP(config-ssid)# authentication open
AP(config-ssid)# wpa-psk ascii 0 SecureWiFi2023
```

```
AP(config)# interface dot11radio 0
AP(config-if)# encryption mode ciphers aes-ccm
```

Verification:

```
AP# show dot11 associations
AP# show interface dot11radio 0
Controller# show wlan summary
```

5.2 EtherChannel (PAgP, LACP)

EtherChannel Concepts

Benefits:

- Increased bandwidth
- Load balancing
- Redundancy
- No spanning tree blocking

Protocols:

- **PAgP**: Cisco proprietary
- **LACP**: IEEE 802.3ad standard

EtherChannel Configuration

LACP Configuration:

```
Switch(config)# interface range fa0/1-2
Switch(config-if-range)# channel-group 1 mode active
Switch(config-if-range)# exit
Switch(config)# interface port-channel 1
Switch(config-if)# switchport mode trunk
```

PAgP Configuration:

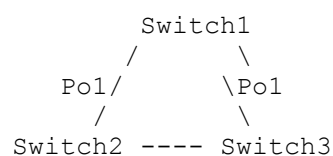
```
Switch(config)# interface range fa0/3-4
Switch(config-if-range)# channel-group 2 mode desirable
Switch(config-if-range)# exit
Switch(config)# interface port-channel 2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

Manual EtherChannel:

```
Switch(config)# interface range gi0/1-2
Switch(config-if-range)# channel-group 3 mode on
Switch(config-if-range)# exit
Switch(config)# interface port-channel 3
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
```

Lab Task 5.2: EtherChannel Implementation

Topology:



Switch1 Configuration:

```
Switch1(config)# interface range fa0/1-2
Switch1(config-if-range)# channel-group 1 mode active
Switch1(config-if-range)# exit
```

```
Switch1(config)# interface range fa0/3-4
Switch1(config-if-range)# channel-group 2 mode active
Switch1(config-if-range)# exit
```

```
Switch1(config)# interface port-channel 1
Switch1(config-if)# switchport mode trunk
```

```
Switch1(config)# interface port-channel 2
Switch1(config-if)# switchport mode trunk
```

Verification:

```
Switch# show etherchannel summary
Switch# show etherchannel port-channel
Switch# show interfaces port-channel 1
Switch# show pagp counters
Switch# show lacp counters
```

Load Balancing Configuration:

```
Switch(config)# port-channel load-balance src-dst-mac
```

5.3 First-Hop Redundancy Protocols (HSRP, VRRP, GLBP)

HSRP (Hot Standby Router Protocol)

HSRP Concepts:

- Cisco proprietary
- Active/Standby model
- Virtual IP and MAC address
- Default priority: 100

HSRP Configuration:

```
Router1(config)# interface gi0/0
Router1(config-if)# ip address 192.168.1.2 255.255.255.0
Router1(config-if)# standby 1 ip 192.168.1.1
Router1(config-if)# standby 1 priority 110
Router1(config-if)# standby 1 preempt
Router1(config-if)# standby 1 track serial 0/0/0
```

```
Router2(config)# interface gi0/0
Router2(config-if)# ip address 192.168.1.3 255.255.255.0
```

```
Router2(config-if)# standby 1 ip 192.168.1.1
Router2(config-if)# standby 1 priority 90
```

VRRP (Virtual Router Redundancy Protocol)

VRRP Configuration:

```
Router1(config)# interface gi0/0
Router1(config-if)# ip address 192.168.1.2 255.255.255.0
Router1(config-if)# vrrp 1 ip 192.168.1.1
Router1(config-if)# vrrp 1 priority 110
Router1(config-if)# vrrp 1 preempt
```

```
Router2(config)# interface gi0/0
Router2(config-if)# ip address 192.168.1.3 255.255.255.0
Router2(config-if)# vrrp 1 ip 192.168.1.1
Router2(config-if)# vrrp 1 priority 90
```

Lab Task 5.3: HSRP Implementation

Topology:

```
PC1 ---- Switch ---- Router1 (Active)
PC2 ---- | ---- Router2 (Standby)
          |
          Default Gateway: 192.168.1.1 (Virtual IP)
```

Router1 (Primary) Configuration:

```
R1(config)# interface gi0/0
R1(config-if)# ip address 192.168.1.10 255.255.255.0
R1(config-if)# standby version 2
R1(config-if)# standby 1 ip 192.168.1.1
R1(config-if)# standby 1 priority 110
R1(config-if)# standby 1 preempt delay minimum 30
R1(config-if)# standby 1 track gi0/1 60
R1(config-if)# standby 1 authentication md5 key-string hsrp123
```

Router2 (Secondary) Configuration:

```
R2(config)# interface gi0/0
R2(config-if)# ip address 192.168.1.20 255.255.255.0
R2(config-if)# standby version 2
R2(config-if)# standby 1 ip 192.168.1.1
R2(config-if)# standby 1 priority 90
R2(config-if)# standby 1 preempt delay minimum 30
R2(config-if)# standby 1 authentication md5 key-string hsrp123
```

Verification:

```
Router# show standby
Router# show standby brief
Router# debug standby events
```

Testing Failover:

1. Shutdown active router's interface
2. Verify standby takes over
3. Check PC connectivity maintained
4. Bring interface back up
5. Verify preemption if configured

5.4 Advanced Switching Features

Port Security

Port Security Configuration:

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security mac-address sticky
Switch(config-if)# switchport port-security violation restrict
```

Violation Actions:

- **Protect:** Drops packets, no notification
- **Restrict:** Drops packets, sends SNMP trap
- **Shutdown:** Disables port (default)

DHCP Snooping

DHCP Snooping Configuration:

```
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan 10,20
Switch(config)# interface fa0/1
Switch(config-if)# ip dhcp snooping trust
Switch(config)# ip dhcp snooping information option
```

Dynamic ARP Inspection (DAI)

DAI Configuration:

```
Switch(config)# ip arp inspection vlan 10
Switch(config)# interface fa0/1
Switch(config-if)# ip arp inspection trust
Switch(config)# ip arp inspection validate src-mac dst-mac ip
```

Lab Task 5.4: Advanced Security Features

Topology:

```
DHCP_Server ---- Trusted_Port [Switch] Untrusted_Ports ---- PCs
                        |
                        Trusted_Port ---- Router
```

Security Configuration:

```
# Enable DHCP Snooping
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan 10

# Configure trusted ports
Switch(config)# interface fa0/24
Switch(config-if)# ip dhcp snooping trust

# Configure port security
Switch(config)# interface fa0/1
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 1
Switch(config-if)# switchport port-security mac-address sticky
Switch(config-if)# switchport port-security violation shutdown

# Enable DAI
Switch(config)# ip arp inspection vlan 10
Switch(config)# interface fa0/24
Switch(config-if)# ip arp inspection trust
```

Verification:

```
Switch# show ip dhcp snooping binding
Switch# show port-security
Switch# show ip arp inspection
```

5.5 Troubleshooting

Common Troubleshooting Commands

Layer 1 (Physical):

```
Router# show interfaces
Router# show controllers serial 0/0/0
Switch# show interfaces status
Switch# show power inline
```

Layer 2 (Data Link):

```
Switch# show mac address-table
Switch# show spanning-tree
Switch# show vlan brief
Switch# show interfaces trunk
```

Layer 3 (Network):

```
Router# show ip route
Router# show ip arp
Router# ping 192.168.1.1
Router# traceroute 192.168.1.1
```

Layer 4+ (Transport/Application):

```
Router# show ip sockets
Router# telnet 192.168.1.1 80
PC> netstat -an
PC> nslookup google.com
```

Systematic Troubleshooting Approach

1. **Define the Problem**
 - What is not working?
 - When did it stop working?
 - What changed recently?
2. **Gather Information**
 - Network topology
 - Error symptoms
 - Affected devices/users
3. **Consider Possibilities**
 - Layer-by-layer analysis
 - Common failure points
 - Recent changes
4. **Create Action Plan**
 - Start with most likely cause
 - Least disruptive tests first
 - Document findings
5. **Implement and Test**
 - Make one change at a time
 - Test after each change
 - Roll back if unsuccessful
6. **Monitor and Document**
 - Verify solution works
 - Document resolution
 - Implement preventive measures

Lab Task 5.5: Comprehensive Network Troubleshooting

Scenario: Users in VLAN 10 cannot access the internet, but VLAN 20 users can.

Troubleshooting Steps:

1. **Verify Physical Layer:**
2. Switch# show interfaces fa0/1
3. Switch# show interfaces status
4. **Check Data Link Layer:**
5. Switch# show vlan brief

```
6. Switch# show interfaces fa0/1 switchport
7. Switch# show mac address-table vlan 10
8. Verify Network Layer:
9. Router# show ip route
10. Router# show ip interface brief
11. PC> ipconfig /all
12. PC> ping 192.168.10.1
13. Test Connectivity:
14. PC> ping 8.8.8.8
15. PC> nslookup google.com
16. Router# show ip nat translations
17. Check ACLs and Policies:
18. Router# show access-lists
19. Router# show ip interface gi0/0
```

Common Issues and Solutions:

Issue	Symptoms	Solution
Wrong VLAN assignment	Can't reach gateway	Verify VLAN configuration
Missing route	Can reach local, not remote	Add static/dynamic routes
ACL blocking	Selective connectivity issues	Review ACL rules
NAT problems	Can't reach internet	Check NAT configuration
DHCP issues	No IP address	Verify DHCP server/relay

Summary and Next Steps

CCNA Exam Topics Coverage

This study guide covers all major CCNA exam topics:

✓ Network Fundamentals (20%)

- OSI and TCP/IP models
- Ethernet concepts
- IPv4/IPv6 addressing and subnetting

✓ Network Access (20%)

- VLANs and trunking
- EtherChannel
- Wireless concepts

✓ IP Connectivity (25%)

- Routing concepts
- Static and dynamic routing
- First-hop redundancy protocols

✓ IP Services (10%)

- NAT/PAT
- DHCP
- NTP, SNMP, Syslog

✓ Security Fundamentals (15%)

- ACLs
- Wireless security
- Port security

✓ Automation and Programmability (10%)

- Network management
- APIs and configuration management

Recommended Lab Practice Schedule

Week 1-2: Foundation

- Complete all Phase 1 labs
- Focus on basic connectivity and addressing

Week 3-4: Switching

- Master VLAN and trunking concepts
- Practice STP scenarios

Week 5-6: Routing

- Static and dynamic routing protocols
- NAT/PAT configuration

Week 7-8: Services and Security

- DHCP, ACLs, and network services
- Wireless configuration

Week 9-10: Advanced Topics

- EtherChannel and HSRP
- Comprehensive troubleshooting

Additional Resources

- **Cisco Packet Tracer:** Essential for lab practice
- **Cisco Documentation:** Latest configuration guides
- **Practice Exams:** Test knowledge retention

- **Video Training:** Supplement hands-on practice

Final Exam Preparation Tips

1. **Hands-on Practice:** Configuration is key
2. **Understand Concepts:** Don't just memorize commands
3. **Troubleshooting Skills:** Practice systematic approach
4. **Time Management:** Know command shortcuts
5. **Documentation:** Keep configuration templates